Electromechanical Maintenance Management System

Software Requirements and Implementation



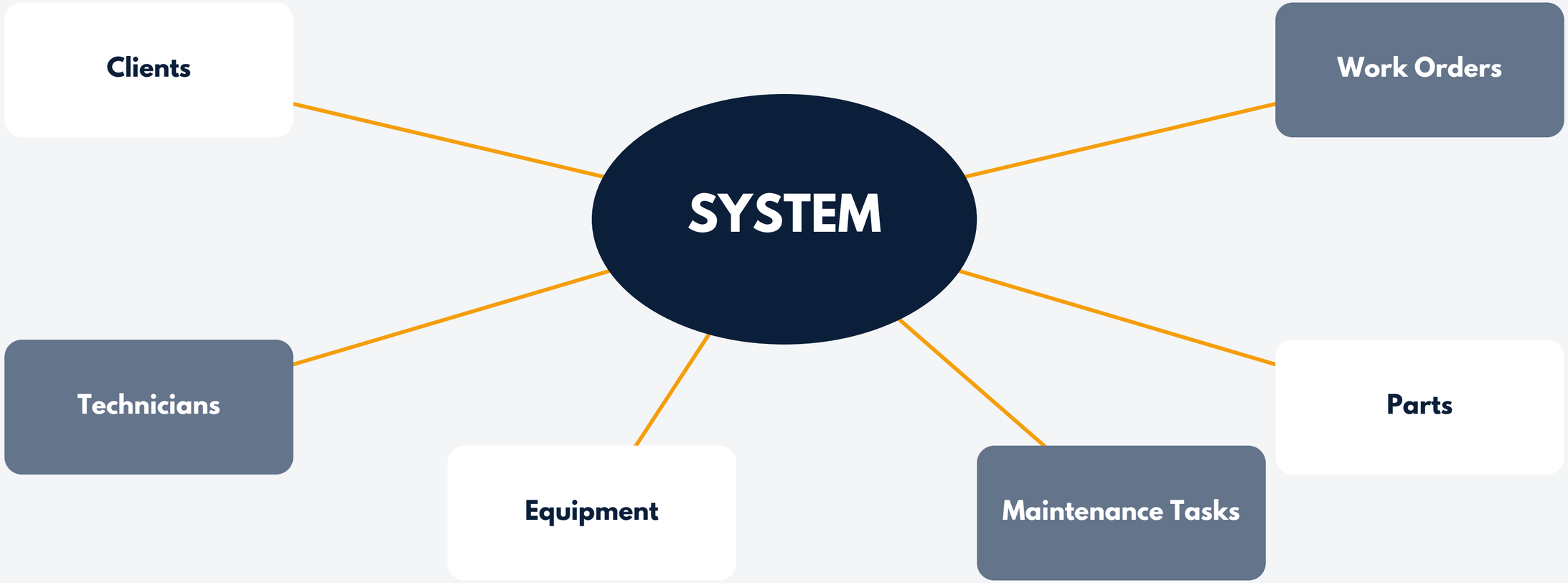
BINARY BOSSES • SOFTWARE ENGINEERING • 2026
Universidad de las Fuerzas Armadas ESPE

Presenters:
Marcos Mamarandi
Mateo Nacimba
Renata Olalla
Instructor: Edison Lascano

One system. Six connected domains.

A centralized solution for electromechanical maintenance management.





Organized maintenance • reduced downtime • better resource use

The SRS aligns scope, people, and proof

Version 1.0 • subject to client feedback from Enrique Nacimba



PURPOSE

Define functional and non-functional requirements. Guide design, implementation, and testing.

SCOPE

Centralize clients, technicians, equipment, orders, schedules, repairs, history, QR codes, reports, and alerts.

REFERENCES
HW04 Definition • HW05 Product Backlog
Binary Bosses, ESPE, 2026

ADMINISTRATOR

Coordinates records, orders, and reports.

TECHNICIAN

Performs maintenance; records tasks and parts.

CLIENT

Consults equipment and maintenance records.

Functional requirements: traceable by design

Every capability maps directly to one product-backlog item.



FR-01 / PB-01

Client Management

Register • update • search

FR-02 / PB-02

Equipment Management

Register • identify • search

FR-03 / PB-03

Technician Management

Skills • availability

FR-04 / PB-04

Work Order Management

Create • assign

FR-05 / PB-05

Preventive Scheduling

Schedule • view

FR-06 / PB-06

Corrective Records

Repairs • notes • parts

FR-07 / PB-07

Maintenance History

Search • display

FR-08 / PB-08

QR Identification

Generate • access

FR-09 / PB-09

Reports and Alerts

Summaries • alerts

Nine requirements define the product boundary

Seven high-priority foundations; two medium-priority extensions.



ID	FEATURE / USER STORY	PRIORITY	ACCEPTANCE CRITERIA
PB01	Client records	HIGH	Save, find, modify
PB02	Equipment + ownership	HIGH	Store and link to client
PB03	Technicians	HIGH	Skills and availability
PB04	Work orders	HIGH	Create and assign
PB05	Preventive schedule	HIGH	Register inspections
PB06	Corrective records	HIGH	Repairs, notes, parts
PB07	Maintenance history	HIGH	Display previous records
PB08	QR identification	MEDIUM	Unique equipment link
PB09	Reports + alerts	MEDIUM	Summaries and reminders

Quality is a requirement—not an afterthought

Five system qualities frame implementation decisions.



NFR-01

PERFORMANCE

Basic operations within 3 seconds.

NFR-02

SECURITY

Username and password login.

NFR-03

USABILITY

Clear menus and simple navigation.

NFR-04

RELIABILITY

Data persists after closing.

NFR-05

MAINTAINABILITY

Separate OO classes per entity.

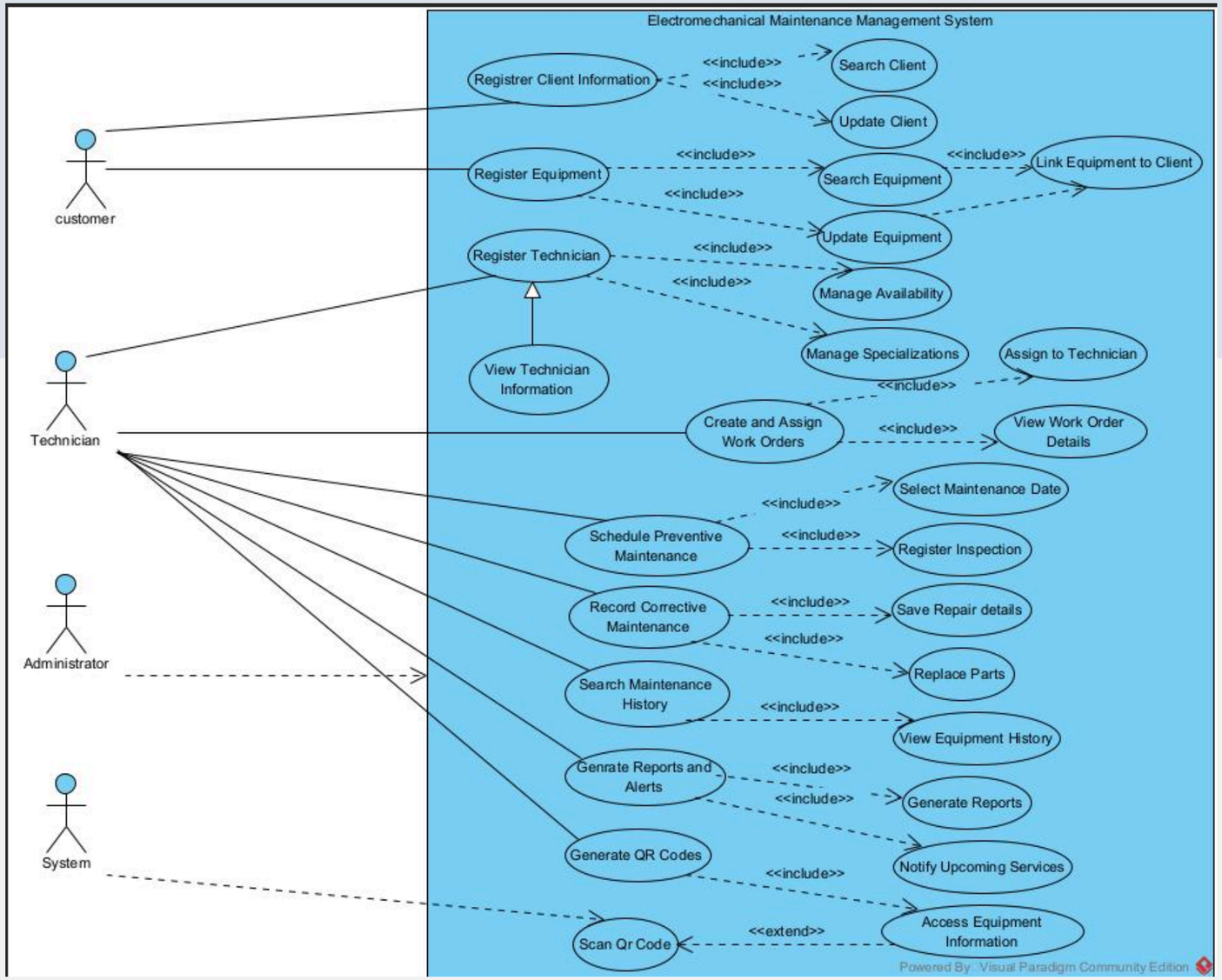
CONSTRAINTS

Object-oriented programming • simple storage • orders require registered equipment and technicians • scope limited to nine backlog features.

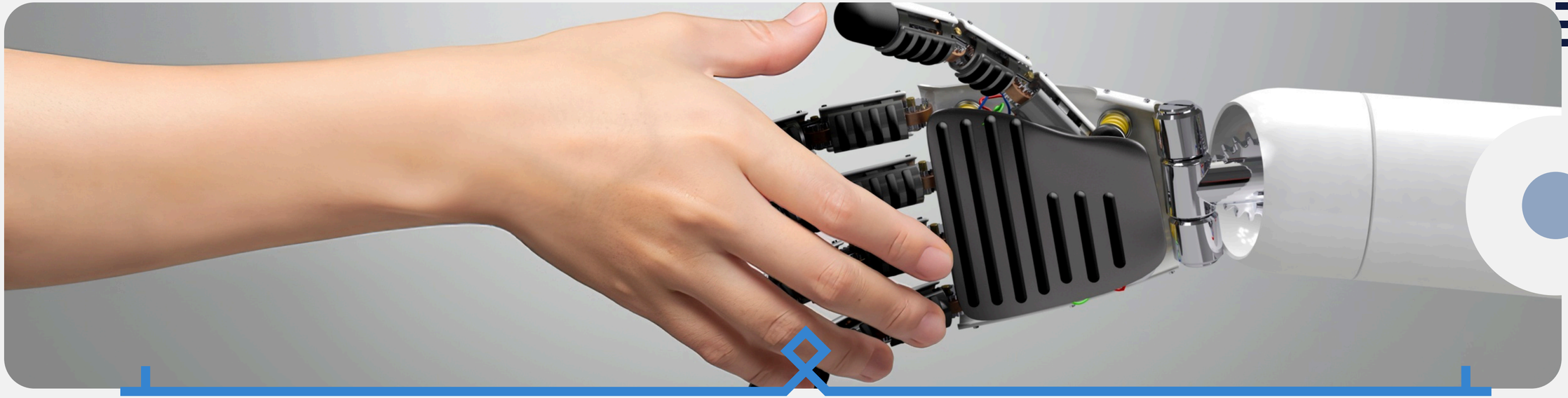
ASSUMPTIONS & DEPENDENCIES

Accurate user data • technicians update records • assignment depends on availability • alerts depend on preventive schedules.

Use Case


Electromechanical Maintenance Management System
customer
Technician
Administrator
System
Registrer Client Information
Search Client
Update Client
Register Equipment
Search Equipment
Link Equipment to Client
Update Equipment
Register Technician
Manage Availability
Manage Specializations
Assign to Technician
View Technician Information
Create and Assign Work Orders
View Work Order Details
Select Maintenance Date
Schedule Preventive Maintenance
Register Inspection
Record Corrective Maintenance
Save Repair details
Replace Parts
Search Maintenance History
View Equipment History
Genrate Reports and Alerts
Generate Reports
Generate QR Codes
Notify Upcoming Services
Access Equipment Information
Scan Qr Code
<<include>>
<<extend>>
Powered By Visual Paradigm Community Edition



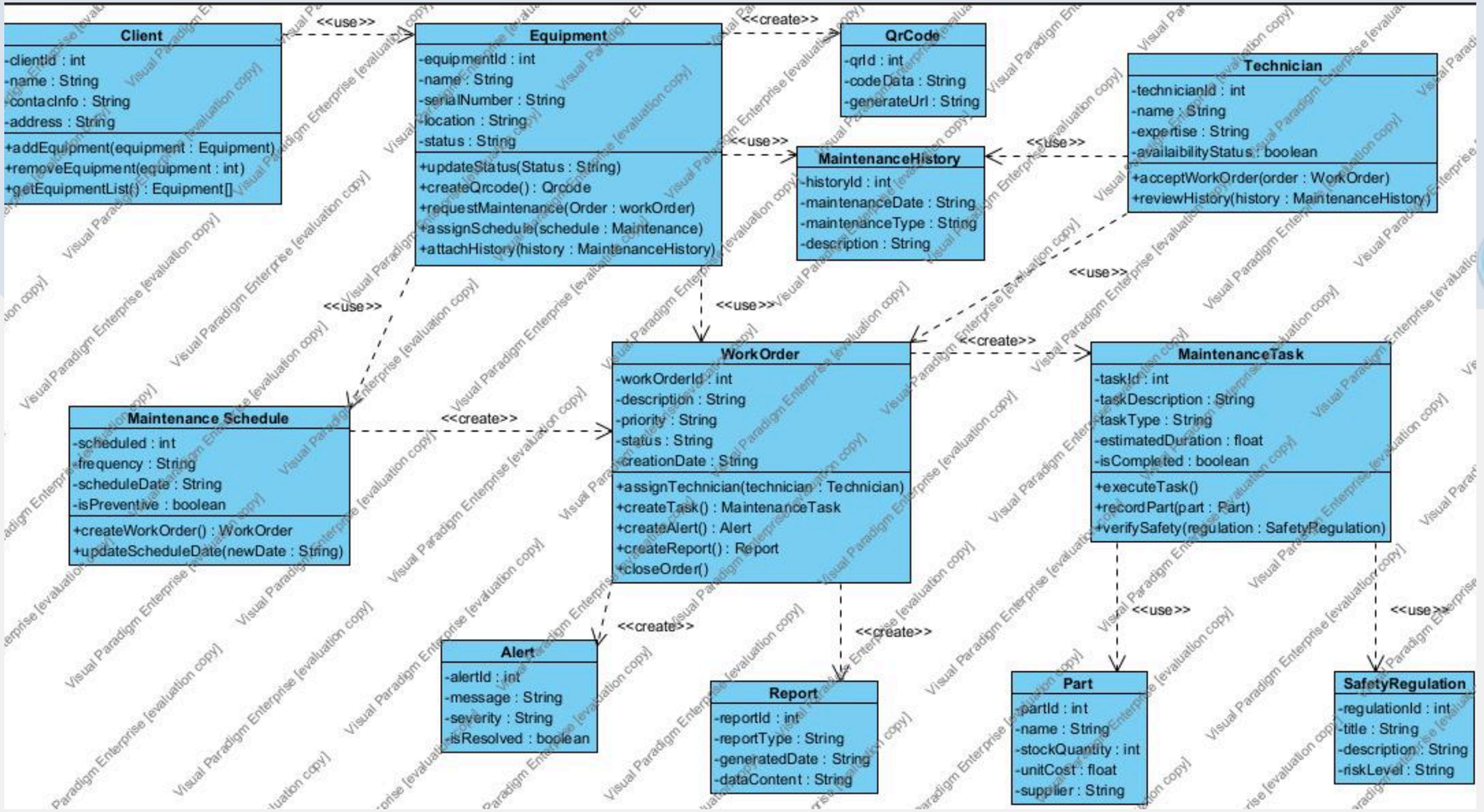








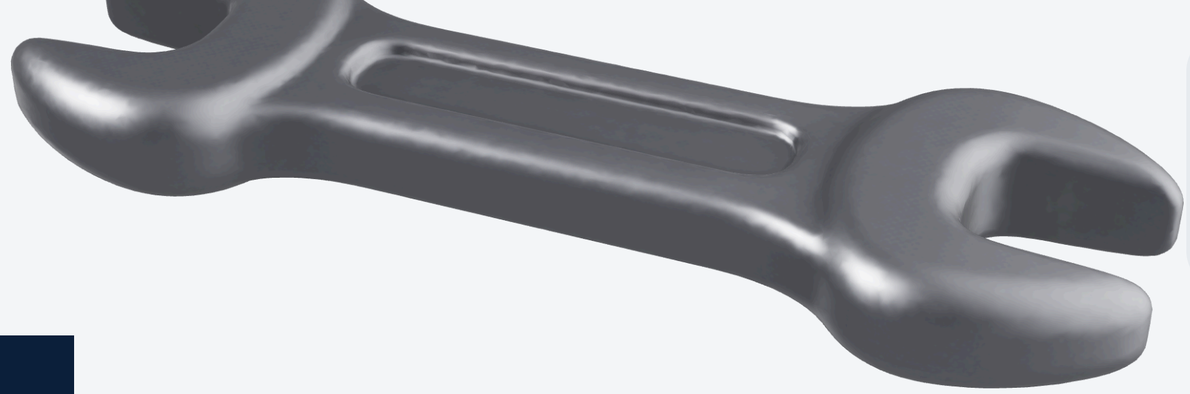
Class Diagram





Project Architecture

Java • Apache NetBeans • Ant • layered separation of concerns



```
PROJECT
├─── src
│    ├─── model   (12 entities)
│    ├─── view    (console menu)
│    └─── controller
├─── test        (JUnit)
└─── testdata    (CSV)
```

controller package:
ec.edu.espe.electromechanicalsystem.controller

model / view package root:
ec.edu.espe.electromechanicalmaintenancemanagementsystem

VIEW

Console menu • 14 options

CONTROLLER

JsonFileManager • JSON persistence

MODEL

12 entity classes • domain behavior

From customer record to field identification

Three model classes establish ownership, equipment state, and QR access.



Client

ATTRIBUTES

clientId: int
name, contactInfo, address: String
equipmentList: List<Equipment>

METHODS

addEquipment(Equipment)
removeEquipment(int)
getEquipmentList(): Equipment[]
getters • setters • toString()

Equipment

ATTRIBUTES

equipmentId: int
name, serialNumber, location,
status: String
qrCode: QrCode
histories • schedules • work orders

METHODS

updateStatus(String)
createQrCode()
requestMaintenanceOrder()
assignSchedule() • attachHistory()

QrCode

ATTRIBUTES

qrId: int
codeData: String
equipment link

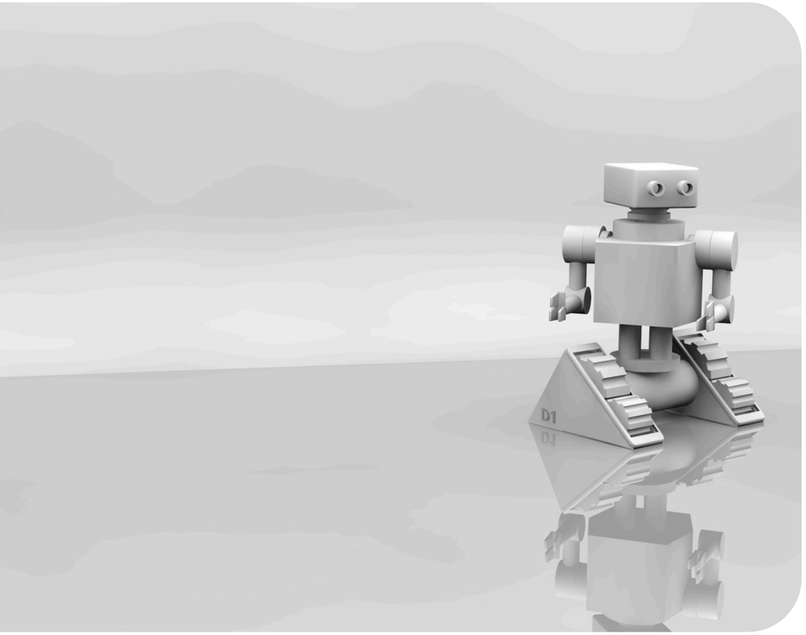
METHODS

generateCodeData()
getters • setters
toString()



01

THE CODE



THANK YOU

